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Migration, aggregation, and philopatry of two nearshore elasmobranch species in the Southern California Bight

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ABSTRACT: Over one-third of elasmobranch fishes (sharks, rays, and skates) are threatened with extinction, mostly due to overfishing, habitat loss, and habitat degradation. Understanding the daily and seasonal movement patterns of these species can inform when and where populations are most susceptible to these threats, but these data are often lacking for nearshore species that are not actively managed. Two such species are the shovelnose guitarfish *Pseudobatos productus* and California bat ray *Myliobatis californica*; this study quantified the broad- and fine-scale movement patterns of these species using passive acoustic telemetry. Twelve guitarfish (10 female, 2 male) were surgically implanted with coded acoustic transmitters at an aggregation site off La Jolla (San Diego County), California, USA, and tracked for 849.5 ± 548.9 d (mean ± SD). Six bat rays (all female) were also implanted with transmitters and tracked for 1143.8 ± 830.9 d. These animals were detected at 187 acoustic receiver stations between Point Conception, California, and San Quintín, Baja California, Mexico. Both species exhibited annual philopatry to La Jolla, especially in July, after traveling as far north as Santa Barbara (221 km away; guitarfish) and San Miguel Island (259 km away; bat rays), California. Based on their movement patterns and known reproductive phenology, we hypothesize that both species utilize the La Jolla aggregation site as a gestating ground and possibly also a mating, pupping, and nursery ground. This site is within a no-take reserve, and we recommend that similar sites also be protected, given the increased susceptibility to anthropogenic stressors when aggregating.

KEY WORDS: Movement patterns · Behavior · Acoustic telemetry · Shovelnose guitarfish · California bat ray · Animal tracking

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1. INTRODUCTION

An estimated 3 billion people worldwide rely on marine fisheries for food (FAO 2014), with small-scale artisanal fisheries comprising more than one-quarter of the global catch (Mathew 2003). Due to this demand, elasmobranch fishes (sharks, rays, and skates) are particularly vulnerable to overexploitation due to

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their slow growth, late maturation, and low fecundity. One-third of elasmobranch species are now threatened with extinction, mostly due to overfishing, habitat loss, and habitat degradation (Dulvy et al. 2021). This is problematic because elasmobranchs serve important roles in ecosystem function. For example, apex predatory sharks may structure biological communities through top-down control, either directly via predation (Baum & Worm 2009) or indirectly via non-consumptive effects (Heithaus et al. 2008). Smaller elasmobranch mesopredators may similarly influence lower trophic level community structure (Baum & Worm 2009). These mesopredators are also vulnerable to overexploitation, especially as commercial fisheries deplete populations of top predators and fish down food webs (Pauly et al. 1998, Pace et al. 1999). Therefore, mesopredatory elasmobranchs must also be carefully managed due to their ecological importance as predators and prey.

To inform these management decisions, understanding the daily and seasonal movement patterns of elasmobranch fishes and how these movements relate to environmental factors can elucidate when and where these species are most vulnerable to anthropogenic stressors, such as fishing pressure and habitat alteration. However, for nearshore mesopredatory species that are not actively managed, movement data are often sparse. Various mesopredator species have been reported to aggregate in certain coastal areas, such as sheltered bays, coves, and estuaries, which have warmer water temperatures. Pregnant females of some nearshore species appear to be attracted to these warm, shallow areas during the summer months, where they may give birth or simply incubate their developing embryos via behavioral thermoregulation (Hopkins & Cech 1994, Salazar-Hermoso & Villavicencio-Garayzar 1999, Hight & Lowe 2007, Jirik & Lowe 2012, Nosal et al. 2014). As a result, these populations are vulnerable to exploitation across multiple generations, particularly by nearshore artisanal and recreational fisheries, and to habitat destruction (Thrush & Dayton 2002, Jennings et al. 2008). Knowledge of how environmental factors, such as water temperature and benthic habitat availability, influence these species' movements is essential for predicting when and where individuals will aggregate and evaluating the effectiveness of marine protected areas (MPAs) and other conservation efforts (Navarro et al. 2016, Matley et al. 2022).

The Southern California Bight (SCB), including the greater Los Angeles area and the San Diego-Tijuana transborder agglomeration, is a region of heavy overlap between anthropogenic stressors such as commercial and recreational fishing, aquatic recreation, and coastal development and is home to several mesopredatory elasmobranch species that are endemic to the west coast of North America. Two such species are the shovelnose guitarfish *Pseudobatos productus* and California bat ray *Myliobatis californica*, which are recreationally and commercially fished, particularly by an underregulated artisanal fishery in Baja California, Mexico (Cartamil et al. 2011, Ramirez-Amaro et al. 2013). The objective of this study was to quantify the broad- and fine-scale movement patterns of shovelnose guitarfish and bat rays in the SCB using passive acoustic telemetry. Specifically, we investigated potential philopatry to a coastal aggregation site, migration patterns to and from the aggregation site, habitat preferences within the aggregation site, and the potential influence of water temperature and photoperiod on these movements.

2. MATERIALS AND METHODS

The waters off La Jolla, California, USA (32.8525° N, 117.2623° W), are frequented by elasmobranch mesopredators, including summer aggregations of leopard sharks *Triakis semifasciata*, round stingrays *Urobatis halleri*, bat rays, and shovelnose guitarfish (Nosal et al. 2013, 2014). The aggregation site is located just inshore of the head (i.e. the lee) of La Jolla Submarine Canyon, where a divergence zone of low wave energy forms, mimicking the warm, calm conditions of a sheltered bay, thus making it an ideal aggregation site for these coastal elasmobranchs (Nosal et al. 2013). There are 4 MPAs in the vicinity of the aggregation site: San Diego-Scripps Coastal State Marine Conservation Area (SMCA), Matlahuayl State Marine Reserve (SMR; no-take), South La Jolla SMCA, and South La Jolla SMR (no-take); these MPAs cover a total of approximately 30 km^2 of nearshore habitat (Fig. 1C). The La Jolla coastline contains a variety of habitat types, including rocky reef, kelp forest, sandflat, submarine canyon, and seagrass bed. Because of its unique combination of having aggregating elasmobranch mesopredators, MPAs, and a diversity of habitat types, La Jolla was chosen as the tagging site for this study.

Shovelnose guitarfish and bat rays were captured from a 5 m skiff at the La Jolla aggregation site in summer and fall of 2014 to 2017 (Table 1) using handlines and barbless circle hooks baited with Pacific mackerel *Scomber japonicus* heads. Hooked individuals were guided into a large scoop net, transferred

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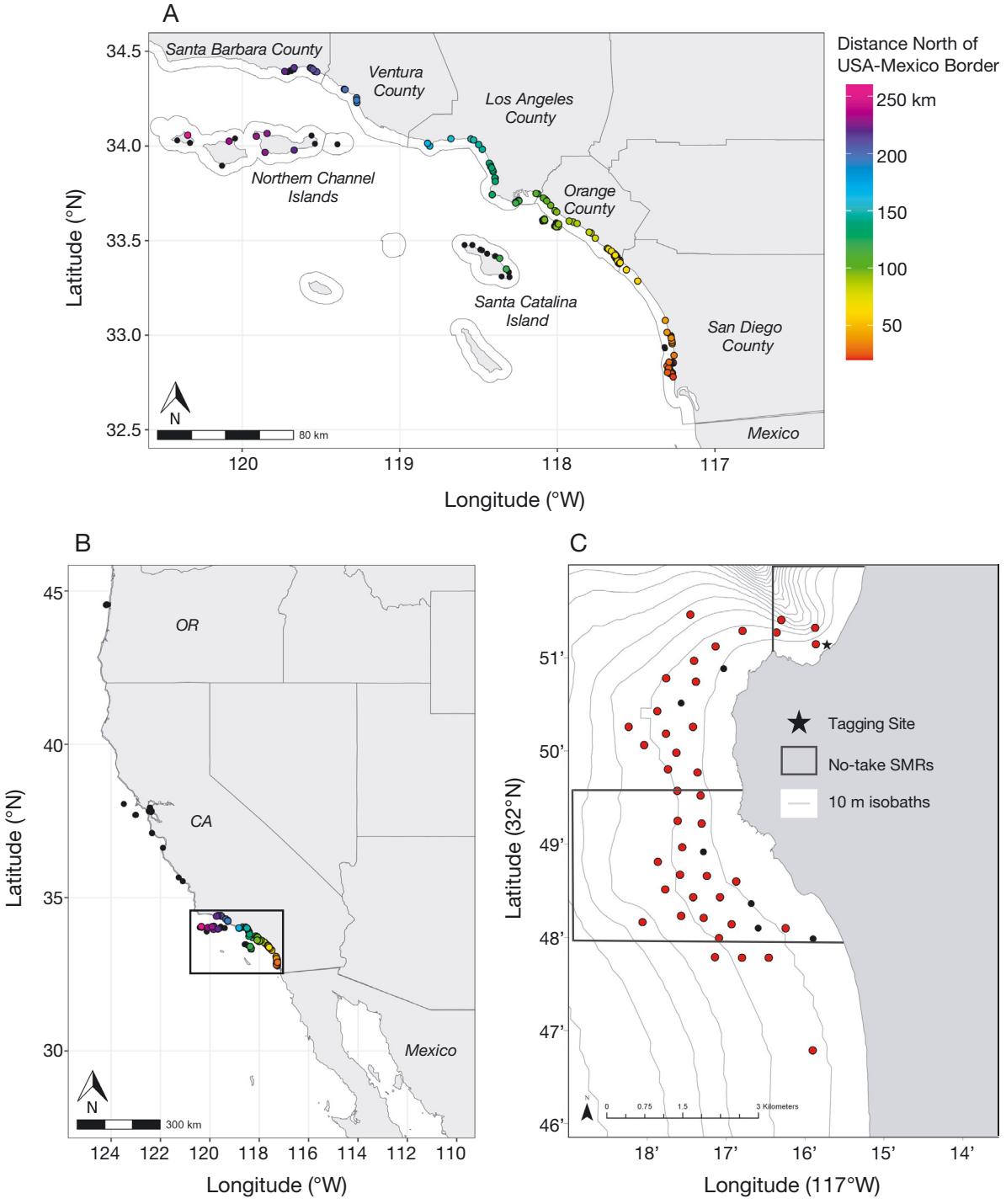

Fig. 1. (A) Coastal array of 187 acoustic receivers that passively tracked shovelnose guitarfish *Pseudobatos productus* and California bat rays *Myliobatis californicus* from June 2014 through July 2020. Black dots represent acoustic receivers known to be deployed during the tracking period that did not detect shovelnose guitarfish or bat rays, and colored dots represent acoustic receivers that did detect transmitter-implanted shovelnose guitarfish or bat rays, colorized by alongshore distance north of the USA–Mexico border (see Section 2 for details). (B) Pacific coast of the USA (OR: Oregon, CA: California). Black and colored dots are as described above. (C) Acoustic receivers off La Jolla, California. Black and colored dots are as described above. Bathymetry is shown as 10 m isobaths (California Fish and Wildlife), and dark gray polygons indicate the boundaries of no-take state marine reserves (SMRs): Matlahuayl SMR (northernmost) and South La Jolla SMR (southernmost)

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Table 1. Summary information for acoustically tracked shovelnose guitarfish (G1–G12) and California bat rays (B13–B18). Known days at liberty are the number of days between the dates of tagging and last detection. Farthest distance is the linear distance between the tagging site and the farthest detecting receiver. For each individual, maximum daily displacement was calculated between sequential detections separated by at least 1 d; the linear distance between those 2 detecting receivers in kilometers was divided by the time in days (including fractions thereof) between sequential detections. Cumulative displacement is the cumulative linear distance between daily mean positions of detecting receivers. Site fidelity index is the proportion of detection hours by 2 receivers southeast of the lee of the La Jolla Submarine Canyon to the total number of detection hours in La Jolla from June to August. DW: disk width; TL: total length

Species	ID	Sex	DW (cm)	TL (cm)	Date tagged (d-mo-yr)	Known days at liberty	Total raw detections	Total detection days	Total detection hours	Farthest distance (km)	Maximum daily displacement (km)	Cumulative displacement (km)	Site fidelity index
Pseudobatos productus	G1	F	44	142	11-Jul-14	404	606	25	82	124	12.6	352	0.26
	G2	F	43	132	11-Jul-14	343	1094	39	232	155	29	298	1
	G3	F	39	123	07-Aug-14	63	214	8	24	178	11.2	168	0
	G4	F	44	131	10-Aug-14	776	803	33	117	286	17.3	786	0.93
	G5	F	47	136	13-Jul-16	91	817	18	93	39	13.1	64	0.92
	G6	M	41	124	14-Jul-16	1505	7974	162	778	252	29.8	2224	0.8
	G7	F	46	140	14-Jul-16	873	316	23	104	192	22.5	199	1
	G8	F	48	147	14-Jul-16	1082	1964	73	286	124	31.8	995	0.98
	G9	F	50	136	15-Jul-16	1445	18 699	200	1498	275	20.9	1182	0.95
	G10	M	42	135	18-Jul-16	1477	10 784	152	1197	68	8.3	565	1
	G11	F	48	146	18-Jul-16	1475	6013	192	1083	273	35.2	2285	0.98
	G12	F	46	131	15-Aug-17	660	29 194	167	1788	283	35	766	1
Myliobatis californicus	B13	F	114	84	17-Jul-14	1804	2098	84	267	316	43.2	3031	0.57
	B14	F	120	87	10-Oct-14	184	385	35	93	72	30.5	209	0
	B15	F	124	99	07-Jul-15	1495	6731	97	608	61	7.7	380	0.55
	B16	F	99	99	08-Jul-15	1881	667	22	116	284	4.2	382	0.05
	B17[a]	F	79	59	01-Jul-16	1497	54 572	592	4187	313	33.5	1764	0.97
	B18	F	117	102	01-Jul-16	2	65	3	9	18	4.6	8	0.11

[a]Immature according to criteria from Martin & Cailliet (1988)

onto the deck of the skiff, and turned ventral side up to induce tonic immobility. Once the hook was removed, each individual was sexed, measured, and surgically implanted with a coded acoustic transmitter (Vemco V16-4H 69 kHz, 158 dB, 120 s average transmission delay, 80–160 s random transmission interval, 2440 d battery life). The transmitter, dipped in povidone iodine, was inserted into the peritoneal cavity via a 3 cm abdominal incision made ~3 cm off the ventral midline. Next, the incision was closed using 1 continuous absorbable suture (Ethicon 2-0 VICRYL) and treated with topical antibiotic ointment (Neosporin). Finally, the animal was righted dorsal side up, externally fitted with a spaghetti identification tag (Floy Tag FIM-96) inserted into the musculature ~3 cm off the dorsal midline, and released. All procedures described above were conducted under University of California, San Diego, Institutional Animal Care and Use Committee Protocol S00080 and California Department of Fish and Wildlife Scientific Collecting Permit 183020007.

Transmitter-implanted shovelnose guitarfish and bat rays were subsequently monitored by a coastal network of underwater acoustic receivers (Vemco VR2Tx, VR2AR, VR2W, VR2C, and VR4-UWM) from Baja California, Mexico, to Oregon, USA, through 31 August 2020 (Fig. 1). Acoustic receivers that most frequently detected our tagged fish were maintained by the Brice Semmens Laboratory at Scripps Institution for Oceanography; the Shark Lab at California State University, Long Beach; and the Channel Islands National Marine Sanctuary. Acoustic detections would have been deemed spurious and removed from the dataset if they did not occur within 1 h (30 times the average transmission delay of 120 s; Pincock 2012) of another detection or if an animal had duplicate detections (e.g.

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detected simultaneously by 2 or more receivers with overlapping detection ranges); however, no such detections were found. All analyses of the spatiotemporal distribution of acoustic detections were performed in R version 4.1.0 (R Core Team 2021) and ArcMap 9.0 (ESRI 2004).

First, given the curvature of the southern California coastline, which complicates interpretations of latitudinal seasonal movement (e.g. movement along the largely north–south oriented coastline of San Diego County would be captured by a change in latitude, whereas movement along the largely east–west oriented coastline of Santa Barbara County would not), the location of each acoustic receiver was paired with its nearest point along the mainland coastline, and its alongshore distance north of the USA–Mexico border was calculated (Fig. 1A). This metric of alongshore distance north of the USA–Mexico border, which was defined as kilometer marker 0, better approximated coastal movement patterns and was used for some analyses below. This metric is appropriate because the majority of receivers were near the mainland coast. Although there were some receivers located around the offshore California Channel Islands, these recorded few detections, and there was little overlap in the metric between coastal receivers and island receivers; therefore, these island receivers were also included in this metric. Additionally, monthly sea surface temperature (SST) data were retrieved via the NOAA multi-scale ultra-high resolution SST analysis (JPL MUR MEaSUREs Project 2015) at 0.01° resolution for the SCB between July 2014 and August 2020. A buffer line was created 5.5 km (3.0 nautical miles) seaward from the mainland coastline, which demarks California state waters, and SST was obtained approximately every 10 km along this buffer line. Then, as above, the location of each SST measurement was paired with its nearest point along the mainland coastline and its distance north of the USA–Mexico border calculated. The alongshore distance north of the USA–Mexico border was then rounded to the nearest km, and any gaps in SST coverage were filled via linear interpolation.

Next, for each species, a generalized linear mixed model (GLMM) was fit to determine the significance and relative influence of environmental cues, or covariates, on coastal movement. The response variable was alongshore distance. Fixed effects considered in model fitting were SST off La Jolla, California (NOAA National Buoy Data Center; Scripps Nearshore Buoy Stn 46254); photoperiod for La Jolla (NOAA Earth Systems Research Laboratories, Global Monitoring Laboratory); and alongshore temperature gradient, which was calculated monthly by plotting remotely sensed SST as a function of alongshore distance north of the USA–Mexico border and retrieving the slope of the best-fit line. Year was included as a random effect, along with individual ID by species nested within year. GLMMs were fit using the lmer() function in the lme4 package in R (Bates et al. 2015). Fixed effects were checked for collinearity using the ggpairs() function in the GGally package (Schloerke et al. 2021). Photoperiod and the alongshore temperature gradient were deemed collinear (correlation = 0.753). For each species, a model including all fixed effects was averaged using the dredge() function in the MuMIn package (Barton 2009), and the best-fit model was chosen based on the Akaike information criterion (AIC) value. Fixed effects from the best-fit model were analyzed in the partR2 package to determine the correlation and strength of each of the environmental cues (Nakagawa & Schielzeth 2013). Inclusive R^2 values were used to quantify each fixed variable's explained variance of the model. Structural coefficients were calculated to represent the correlation of each covariate with alongshore distance, where −1 represented a strong negative correlation and 1 represented a strong positive correlation. Beta weights estimated the relative significance of the environmental drivers on alongshore distance, where CIs that overlapped with 0 were not significant. The strength of influence of the driver on alongshore distance was also determined by the beta weights, where values further from 0 indicated stronger influence.

To quantify philopatry, a residency index (RI) was created for 8 regions: La Jolla (Fig. 1C), San Diego County (excluding La Jolla), Orange County, Los Angeles County, Ventura County, Santa Barbara County, Santa Catalina Island, and the Northern Channel Islands (Fig. 1A). This RI was created following the methods of Anderson et al. (2021):

$$\mathrm{RI} = \frac{\text{days detected in a particular region}}{\text{total number of detection days}} \quad (1)$$

The RI was calculated regionally for each individual and then averaged by species, where an RI of 0 indicated low residence and an RI of 1 indicated high residence.

To quantify space use around La Jolla, California, and to minimize bias caused by the inevitable variability in the ability of receivers to detect transmitters, raw detections were reduced to hours detected for each individual at each receiver. That is, whether an individual was detected 2 or 20 times at a receiver within an hourly bin, that counted as 1 detection hour

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at that receiver. From these detection hours, daily mean positions were calculated for each individual, equivalent to the short-term center of activity prescribed by Simpfendorfer et al. (2002) but with a period of 24 h. From these daily mean positions, and given the irregular shape of the coastline, lattice density estimations were computed for species using the latticeDensity package in R, along with the 95 and 50% activity space polygons (Barry & McIntyre 2011). A manual smoothing parameter (k) of 10 was visually selected over the empirical unbiased cross-validation due to the irregular dispersion of daily mean positions (Kneebone et al. 2012). To supplement the lattice density estimation and to determine site fidelity to the lee of the La Jolla Submarine Canyon during the summer months (when SST was warmest in La Jolla), a site fidelity index for each individual was created. This site fidelity index was calculated as the proportion of the total number of detection hours in La Jolla in the summer months that occurred at the 2 receivers in the lee of the La Jolla Submarine Canyon (Fig. 1C).

Lastly, for each species, paired Wilcoxon rank sum tests were performed on detection hours to test whether there was a diel difference in when individuals were detected generally by the array of receivers off La Jolla and specifically by the 2 receivers in the lee of La Jolla Submarine Canyon (the known aggregation site). Paired Wilcoxon rank sum tests were also used by species to test whether there was a diel difference in the mean distance from land of detecting receivers.

Habitat selection was calculated by collocating daily mean positions to substrate data, categorized as hard and soft substrate (California State Coastal Conservancy and San Diego Association of Governments 2000), by species, using a habitat selection index (HSI; Meese & Lowe 2019):

$$\text{HSI} = \frac{\text{proportion of daily mean positions in a particular substrate}}{\text{proportion of area covered by that particular substrate}} \quad (2)$$

HSI values less than 1 indicated an aversion to that substrate, whereas values greater than 1 indicated an affinity to that substrate. Substrate type (hard and soft) was determined for each daily mean position in ArcMap, with the available area defined as the water area inside the minimum convex polygon (water area as opposed to land area that happened to fall within the minimum convex polygon) encompassing the circular detection ranges (estimated to be 300 m radius after Nosal et al. 2014) of all the acoustic receivers off La Jolla. For each species, a chi-squared test was used to determine whether the proportion of detections within hard and soft substrate was significantly different to the proportion of hard and soft substrate in the available area, and a Welch's 2-sample t-test was used to test whether each species exhibited a significant affinity or aversion to hard and soft substrate.

3. RESULTS

Twelve shovelnose guitarfish were tagged in July and August of 2014 to 2017 (G1–G12; Table 1); 10 were mature females and 2 were mature males, according to the criteria of Timmons & Bray (1997). Six bat rays were tagged in July, August, and October of 2014 to 2016 (B13–B18; Table 1); 5 were mature females and 1 was an immature female, according to the criteria of Martin & Cailliet (1988). For shovelnose guitarfish, mean total length (TL) ± SD was 135.25 ± 7.66 cm, and mean disk width (DW) was 44.79 ± 3.33 cm. For bat rays, mean TL was 88.33 ± 17.66 cm and mean DW was 108.83 ± 16.94 cm. Days at liberty averaged 849.5 ± 548.9 d for shovelnose guitarfish and 1143.8 ± 830.9 d for bat rays; there were no known recaptures (Table 1). Shovelnose guitarfish logged an average of 6540 ± 9104 detections (91 ± 76 detection days), and bat rays logged an average of 10 753 ± 21 609 detections (139 ± 225 detection days; Table 1). Shovelnose guitarfish averaged a cumulative displacement of 823.7 ± 752.1 km, with a maximum daily displacement of 22.2 ± 9.7 km, and bat rays averaged a cumulative daily displacement of 962.3 ± 1189.6 km, with a maximum daily displacement of 20.6 ± 17.1 km (Table 1).

3.1. Migration and philopatry

Both shovelnose guitarfish and bat rays exhibited an annual summer return to La Jolla (i.e. annual philopatry), particularly in July and August (Figs. 2 & 3). Both shovelnose guitarfish and bat rays exhibited high RIs to La Jolla (0.55 ± 0.35 and 0.50 ± 0.41, respectively) and low RIs to other regions (Table 2). Four tagged shovelnose guitarfish (33% of tagged individuals) and 3 tagged bat rays (50% of tagged individuals) returned to La Jolla 3 (G8), 4 (G9, G10, G11, B15, B17), or 5 (B13) consecutive years after tagging (Fig. 2). After summer, most shovelnose guitarfish migrated north along the mainland coast and returned south to La Jolla the following summer.

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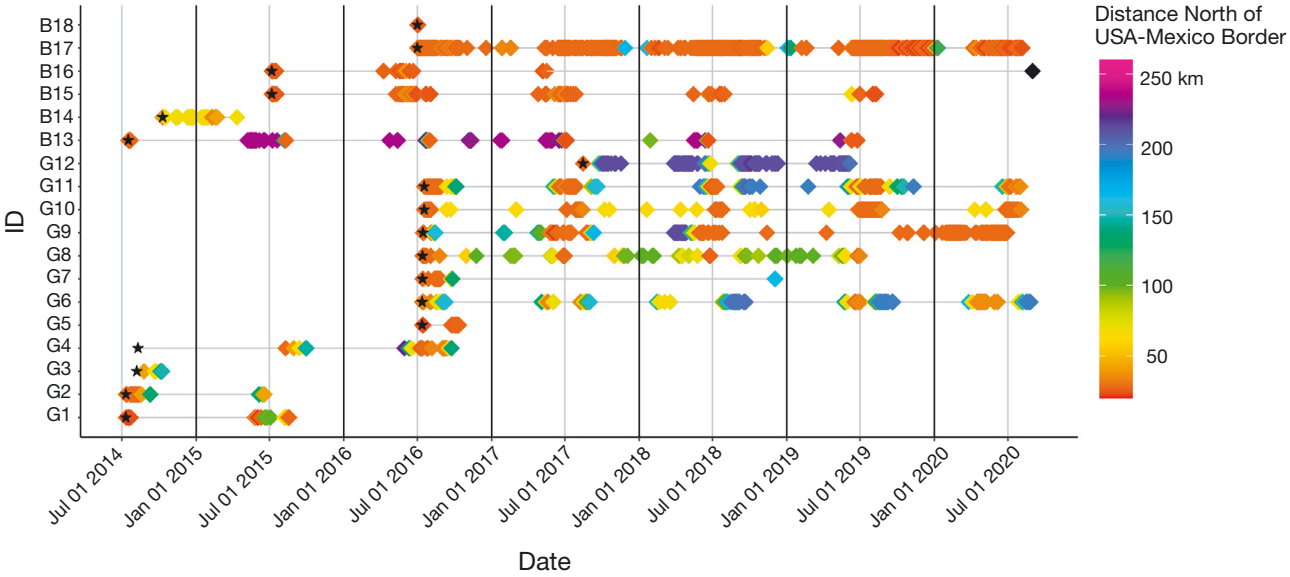

Fig. 2. Acoustic detections of transmitter-implanted shovelnose guitarfish (G1–G12) and California bat rays (B13–B18). Diamonds represent days on which shovelnose guitarfish and bat rays were detected by acoustic receivers, colorized by daily mean alongshore distance north of the USA–Mexico border (see Section 2 for details). The black diamond indicates the day that bat ray B16 was detected by an acoustic receiver off San Quintín, Baja California, Mexico. Stars indicate the time of tagging, and horizontal gray lines represent known time at liberty. Vertical black lines indicate a new calendar year, and vertical grey lines indicate 1 July of each year

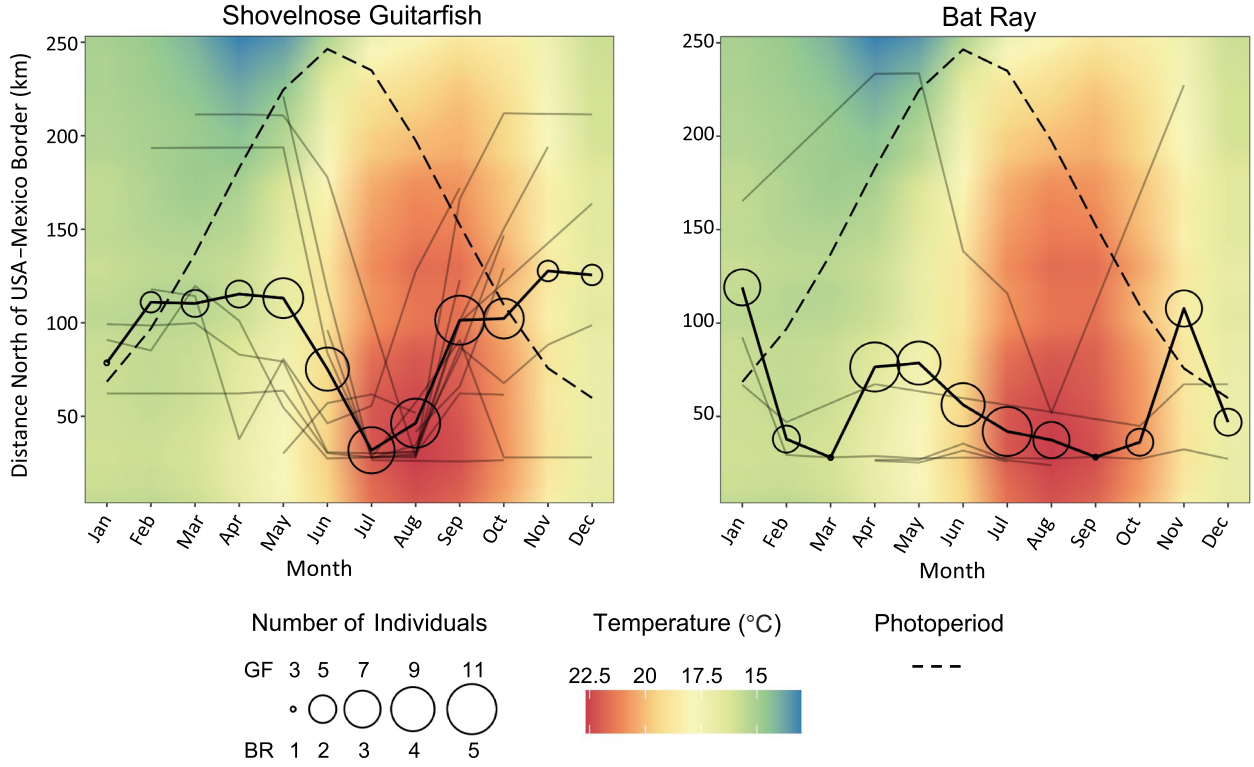

Fig. 3. Monthly mean alongshore distance north of the USA–Mexico border (see Section 2 for details) of shovelnose guitarfish (GF) and California bat rays (BR), shown as solid black lines; the size of open circles along black lines indicates the number of unique individuals detected in that calendar month. Gray lines represent monthly mean alongshore distance by individuals of each species. Dashed black lines represent the mean monthly photoperiod for La Jolla, California. Background raster is colorized by monthly mean sea surface temperature along a buffer line created 5.5 km (3.0 nautical miles) seaward from the mainland coast (see Section 2 for details)

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Table 2. Mean ± SD residency index (RI) of shovelnose guitarfish and California bat rays by region, calculated as the number of days detected in a particular region divided by the total number of days detected anywhere. SD: San Diego County; SB: Santa Barbara County; LA: Los Angeles County

Region	Shovelnose guitarfish	Bat ray
La Jolla (SD)	0.55 ± 0.35	0.50 ± 0.41
San Diego County (except La Jolla)	0.10 ± 0.14	0.11 ± 0.16
Orange County	0.09 ± 0.11	0.16 ± 0.26
Los Angeles County	0.06 ± 0.07	0.10 ± 0.20
Ventura County	0.03 ± 0.07	0.03 ± 0.12
Santa Barbara County	0.18 ± 0.37	0.02 ± 0.10
Northern Channel Islands (SB)	0	0.08 ± 0.2
Santa Catalina Island (LA)	0	0.0006 ± 0.004

Except for B13, bat rays were rarely detected outside of La Jolla during non-summer months; those few detections were confined to receivers off Orange and Los Angeles counties (Fig. 2).

Shovelnose guitarfish were sequentially detected along the mainland coast as far north as Santa Barbara County, California, and as far south as San Diego County, California, at 141 acoustic receiver stations; however, no shovelnose guitarfish were detected at any receivers around the California Channel Islands. Bat rays were detected at 85 acoustic receiver stations along the mainland coast between San Diego and Los Angeles counties (Fig. 2). Although no bat rays were detected along the mainland coast of Ventura or Santa Barbara counties, 2 bat rays (B13 and B17) were detected around the Northern Channel Islands (Fig. 2). Additionally, 1 bat ray (B16) was detected as far south as San Quintín, Baja California, Mexico, by a receiver maintained by the Sosa-Nishizaki Laboratory at Centro de Investigación Científica y de Educación Superior de Ensenada. No individuals of either species were detected north of Point Conception, despite widespread receiver coverage from San Luis Obispo County, California, through Oregon (Nosal et al. 2021).

For shovelnose guitarfish, models including both SST and photoperiod best fit the data, as indicated by the lowest AIC value. For bat rays, models including photoperiod and the alongshore temperature gradient best fit the data, as indicated by the lowest AIC value; however, due to collinearity, this model was not selected. Instead, the second best-fit model, which included SST and alongshore temperature gradient, was selected (ΔAIC = 5.36). For shovelnose guitarfish, alongshore distance north of the USA–Mexico border was most strongly influenced by photoperiod, with a negative correlation (GLMM; Table 3). SST was also a significant cue of alongshore distance, with a negative correlation (Table 3). For bat rays, the alongshore temperature gradient was the strongest cue of alongshore distance, and SST had a non-significant correlation (Table 3).

3.2. Aggregation behavior

Before and during this study, we observed high seasonal densities of guitarfish and bat rays while fishing and snorkeling in the lee of La Jolla Submarine Canyon, suggesting this area is an aggregation site for not only leopard sharks (Nosal et al. 2013, 2014) but also other elasmobranch species. Aerial drone footage recorded near the end of this study provided additional evidence for this area being an aggregation site for guitarfish and bat rays (see Videos S1 & S2 in the Supplement at www.int-res.com/articles/suppl/m724p111_supp/). Both species had high to moderate site fidelity indices during summer months (mean shovelnose guitarfish = 0.82 ± 0.33, mean bat ray = 0.38 ± 0.38; Table 1), and 94.6% of all La Jolla detections were recorded by acoustic receivers within the 2 no-take SMRs. Bat rays had a larger 95% activity space (area = 9.04 km^2) and 50% activity space (area = 0.22 km^2) than shovelnose guitarfish (95% activity space area = 1.53 km^2; 50% activity space area = 0.14 km^2; Fig. 4). Shovelnose guitarfish exhibited a non-random affinity to soft substrate (HSI = 2.86; 95% CI: 2.35–3.37; 1-sample *t*-test, $p < 0.01$) and a significant aversion to hard substrate (HSI = 0.19; 95% CI: 0–0.41; 1-sample *t*-test, $p < 0.01$; Fig. 4C). These HSI values were significantly different from each other (Welch's 2-sample *t*-test, $p < 0.01$). The proportion of shovelnose guitarfish detections over a particular substratum was significantly different from the proportion of that substratum available (chi-squared test; $\chi^2 = 996.11$, df = 1, $p < 0.01$). By comparison, bat rays showed only a slight affinity to soft substratum (HSI = 1.39; 95% CI: 0.46–2.32; Fig. 4C) and a slight aversion to hard substratum (HSI = 0.83; 95% CI: 0.43–1.23; Fig. 4C). However, these HSI values were not statistically significant (1-sample *t*-test, $p = 0.448$) nor were they significantly different from each other (Welch's 2-sample *t*-test, $p = 0.315$). Lastly, the proportion of bat ray detections over a particular substratum was significantly different from the proportion of that substratum available (chi-squared test; $\chi^2 = 892.56$, df = 1, $p < 0.01$).

Of detections made by acoustic receivers off La Jolla (Fig. 1C), the proportion of shovelnose guitarfish

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Table 3. General linear mixed models fitted for shovelnose guitarfish and California bat rays to model the environmental covariates sea surface temperature (SST) off La Jolla, photoperiod, and alongshore temperature gradient (see Section 2 for details) of alongshore distance north of the USA–Mexico border, with individuals nested in year as a random effect (see Section 2 for details). Inclusive R^2 represents the variance that each covariate explains. Structural coefficients represent a fixed effect's correlation, irrespective of other fixed effects. Beta weights with CIs that do not overlap 0 represent significance of that environmental covariate on the predictor variable, and relative beta weights represent strength of correlation with the overall model prediction. AIC: Akaike information criterion

Environmental covariate	Regression coefficient (unstandardized)	Inclusive R^2	Structure coefficient	Beta weight
(a) Guitarfish alongshore distance ~ La Jolla SST + photoperiod + (1 | yr/ID)				
Intercept = 358.021 ± 20.476				
AIC = 10783.66				
Adjusted R^2 = 0.679 (CI: 0.627–0.749)				
La Jolla SST	−2.715 ± 0.628	0.254 (0.208–0.362)	−0.611 (−0.720 to −0.546)	−0.096 (−0.178 to −0.088)
Photoperiod	−16.994 ± 1.291	0.615 (0.508–0.680)	−0.952 (−0.973 to −0.896)	−0.249 (−0.324 to −0.235)
(b) Bat ray alongshore distance ~ La Jolla SST + alongshore temperature gradient + (1 | yr/ID)				
Intercept = 56.711 ± 6.415				
AIC = 1873.01				
Adjusted R^2 = 0.611 (CI: 0.561–0.716)				
La Jolla SST	−0.595 ± 0.295	0.34 (0.163–0.467)	−0.754 (−0.815 to −0.521)	−0.090 (−0.088 to 0.001)
Alongshore temperature gradient	−642.205 ± 153.202	0.554 (0.540–0.621)	−0.952 (−0.996 to −0.917)	−0.194 (−0.223 to −0.102)

detections during the day was significantly higher than that at night (paired Wilcoxon rank sum test; v = 61, p = 0.009). However, no diel pattern was apparent in detections of bat rays (paired Wilcoxon rank sum test; v = 16, p = 0.312). Of detections made by only the 2 acoustic receivers in the lee of La Jolla Submarine Canyon (Fig. 1C), the proportion of shovelnose guitarfish detections made during the day was significantly higher than that at night (paired Wilcoxon rank sum test; v = 66, p = 0.003). A similar, but not statistically different, trend was detected for bat rays (paired Wilcoxon rank sum test; v = 15, p = 0.057). Neither shovelnose guitarfish nor bat rays exhibited a diel difference in the distance of detecting receivers from shore (paired Wilcoxon rank sum test; shovelnose guitarfish: v = 14, p = 0.193; bat ray: v = 4, p = 0.437).

4. DISCUSSION

This study represents the longest tracking period (6 yr) and the largest tracking scale (straight line distance between most distant detecting receivers = 606 km) for either shovelnose guitarfish or California bat rays and, to our knowledge, is the first to track the latter with passive acoustic telemetry. Both species exhibited strong annual philopatry (up to 5 consecutive summer returns, mean RI for both species > 0.5) and high site fidelity (mean site fidelity for both species > 0.4) to the lee of the La Jolla Submarine Canyon off La Jolla, California, a known summer elasmobranch aggregation site (Nosal et al. 2013, 2014, 2021). During the autumn months, both species generally migrated north and west along the southern California coast, with shovelnose guitarfish detected as far north as Santa Barbara, California, and bat rays detected as far north as the Northern Channel Islands. Previous studies on shovelnose guitarfish and bat ray movement patterns and habitat association had been limited to shorter monitoring periods (<1 yr) and enclosed bays in California, including Tomales Bay in the San Francisco Bay area (Matern et al. 2000, Klimley et al. 2005), Bolsa Chica Ecological Reserve (Orange County; Farrugia et al. 2011), Huntington Beach Wetlands (Orange County; Freedman et al. 2015), and Big Fisherman's Cove on Santa Catalina Island (Meese & Lowe 2019). These results, along with an observation of a bat ray detected off San Quintín, Baja California, Mexico, represent the farthest seasonal movements reported for either species.

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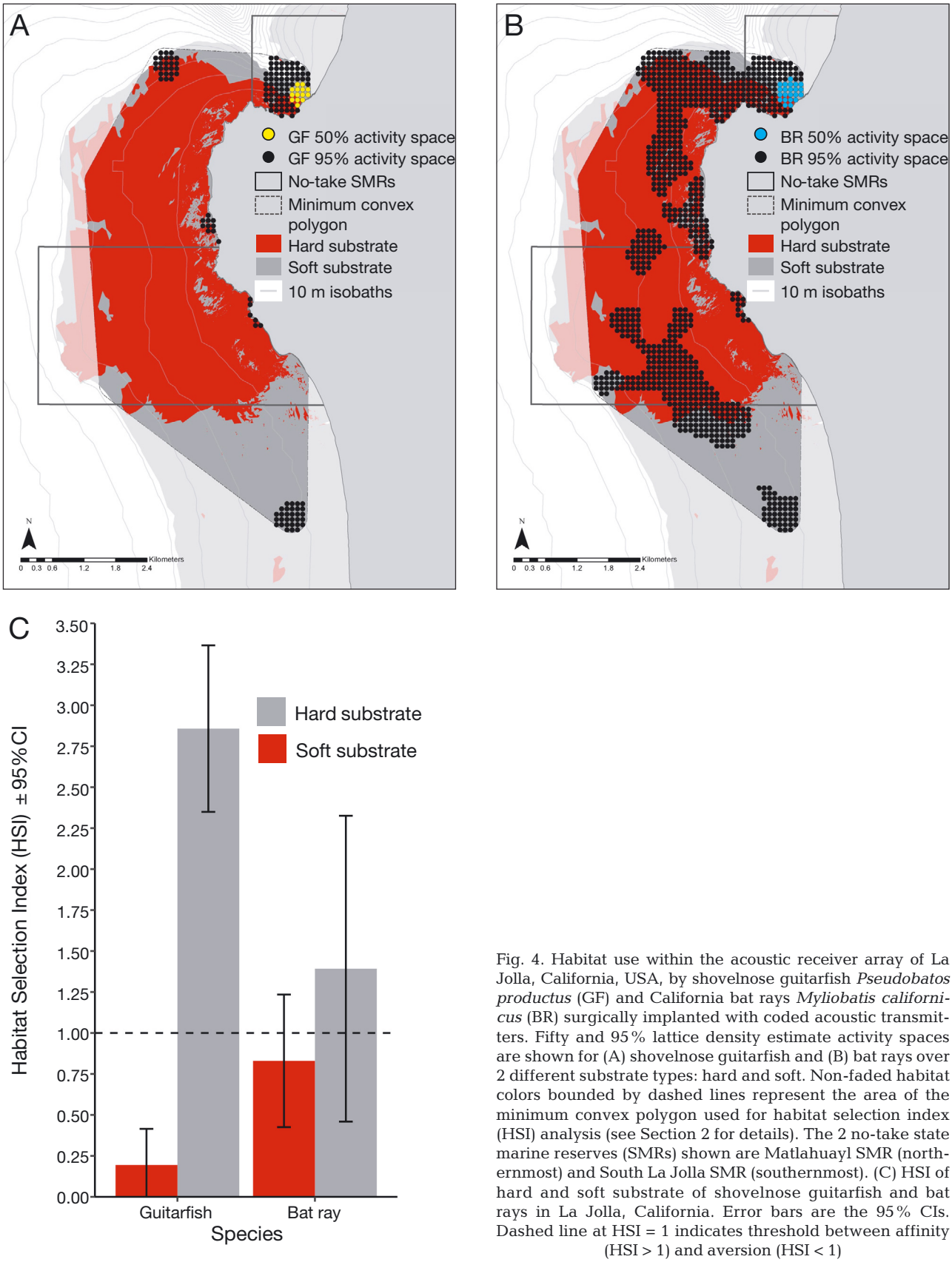

Fig. 4. Habitat use within the acoustic receiver array of La Jolla, California, USA, by shovelnose guitarfish *Pseudobatos productus* (GF) and California bat rays *Myliobatis californicus* (BR) surgically implanted with coded acoustic transmitters. Fifty and 95% lattice density estimate activity spaces are shown for (A) shovelnose guitarfish and (B) bat rays over 2 different substrate types: hard and soft. Non-faded habitat colors bounded by dashed lines represent the area of the minimum convex polygon used for habitat selection index (HSI) analysis (see Section 2 for details). The 2 no-take state marine reserves (SMRs) shown are Matlahuayl SMR (northernmost) and South La Jolla SMR (southernmost). (C) HSI of hard and soft substrate of shovelnose guitarfish and bat rays in La Jolla, California. Error bars are the 95% CIs. Dashed line at HSI = 1 indicates threshold between affinity (HSI > 1) and aversion (HSI < 1)

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4.1. Migration behavior and annual philopatry to La Jolla, California

Both shovelnose guitarfish and bat rays displayed higher residency to La Jolla, California, compared to other regions, with about half of detection days being in La Jolla compared to the 7 other regions. Both species also exhibited annual summer philopatry, generally returning during the months of July and August. This summer aggregation behavior is consistent with artisanal fishery data for shovelnose guitarfish off Baja California Sur, Mexico, which show higher catch per unit effort in summer months (Salazar-Hermoso & Villavicencio-Garayzar 1999). Similarly, elasmobranch derby capture data for both species in Elkhorn Slough and for bat rays in Humboldt Bay also showed increased catches during these months (Talent 1985, Gray et al. 1997, Carlisle et al. 2007).

Reproduction is a likely explanation for both species' return to La Jolla, given their annual reproductive cycles (Martin & Cailliet 1988, Márquez-Farías 2007). Understanding the movement patterns of males via additional tagging and tracking would be useful in assessing whether the La Jolla aggregation site functions as a mating ground. However, in the Gulf of California, mating in shovelnose guitarfish occurred in June and July (Márquez-Farías 2007), which is when tagged shovelnose guitarfish were present in La Jolla in this study, including the 2 males tagged. Additionally, shovelnose guitarfish mating behavior was observed in August 2020 via aerial drone footage over La Jolla (Video S3). However, the skew towards females in our catches and tagged individuals suggests that these aggregations may be dominated by females.

Juvenile shovelnose guitarfish were not observed or captured off La Jolla during our study, so it seems unlikely that La Jolla functions as a major pupping or nursery ground for this species. Instead, female shovelnose guitarfish may move north to give birth, likely in suspected nursery areas between Huntington Beach (Orange County) and Long Beach (Los Angeles County), California, including the Bolsa Chica Ecological Reserve (Orange County; Martin & Cailliet 1988, Farrugia et al. 2011). Bat rays, on the other hand, may use La Jolla as a pupping and nursery ground. In July 2015 and 2016, we observed the fins of term pups protruding from the cloaca of 2 of the mature females we tagged in this study, suggesting parturition was imminent. Additionally, drone footage in August 2019 over La Jolla has also captured aggregations of hundreds of bat rays of various sizes, including juveniles (Video S2). Although this study cannot determine if this site fulfills the criteria for an elasmobranch nursery ground, these observations do suggest that juvenile bat rays may be aggregating in La Jolla (Heupel et al. 2007).

The movements of tagged shovelnose guitarfish and bat rays in this study were within their previously reported species ranges: Baja California, Mexico, to San Francisco, California, USA, for shovelnose guitarfish (Farrugia et al. 2016) and to Oregon, USA, for bat rays (van Hees et al. 2015). Previous genetics work on the Pacific coast population of shovelnose guitarfish identified 2 geographic barriers to gene flow: (1) the San Pedro Channel, and (2) Punta Eugenia, Mexico (Meyer 2020). As a strictly benthic species, it is unlikely that shovelnose guitarfish would cross the deep San Pedro Channel separating the California Channel Islands from the mainland, and the absence of shovelnose guitarfish detections from around the California Channel Islands is consistent with genetically isolated island populations, as seen in round rays (Plank et al. 2010). However, given the lesser degree of acoustic receiver coverage in Baja California, Mexico, we could not measure the extent to which shovelnose guitarfish travel south of the USA–Mexico border. On the other hand, the absence of detections from north of Point Conception, which had fairly robust coastal acoustic receiver coverage, is consistent with the well-known biogeographic boundary of Point Conception that separates warmer southern and colder northern water masses along the Pacific coast (Stepien et al. 1991). This boundary has previously been characterized as a barrier to gene flow in other mesopredatory elasmobranchs, including brown smoothhound and leopard sharks (Barker et al. 2015, Chabot et al. 2015).

While the genetic structure of bat rays has not been investigated, the absence of detections north of Point Conception may suggest that this bioregion break represents a barrier to gene flow for this species as well. However, unlike shovelnose guitarfish, the deep channel separating the Northern Channel Islands does not seem to be a barrier to the movement of bat rays, with 2 individuals moving between the mainland and the Northern Channel Islands. Bat ray B13, in particular, made routine movements between La Jolla and the Northern Channel Islands (Fig. 1). During aerial surveys in the autumn, large groups of adult bat rays have been seen over 3 km offshore swimming at the surface in a uniform direction, presumably migrating north (C. G. Lowe pers. comm.). Additionally, given that bat ray B16 was detected off San Quintín, Mexico, this may suggest a single subpopulation ranging from Point Conception, California, to at least Punta Eugenia, Mexico. Despite the an-

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nual summer return to La Jolla, the absence of detections anywhere north of San Diego for B15 and B16 suggests that these individuals may have frequented Baja California, where there was lesser acoustic receiver coverage. Similar long-term wide-range passive acoustic tracking studies should be conducted on shovelnose guitarfish and bat rays north of Point Conception as well as south of the USA–Mexico border to better understand their seasonal movement patterns and compare them to other subpopulations.

Daylight was an important predictor of shovelnose guitarfish movement on a seasonal timescale. Photoperiod, independent of water temperature, was a significant predictor of seasonal presence in La Jolla. This study is among the few to demonstrate the importance of photoperiod on elasmobranch movement and only the second study to do so in the SCB. In Nosal et al. (2014), photoperiod, independent of water temperature, was found to be a strong predictor of male leopard sharks returning to an aggregation site off Del Mar, California. Shovelnose guitarfish may use increasing photoperiod, which precedes increasing water temperature, as a cue to return to gestation grounds off La Jolla. Meanwhile, the alongshore temperature gradient of the region, independent of photoperiod, is an important predictor of bat ray movement on a seasonal timescale. This could be an indicator that is concurrent with increasing SST, stimulating individuals to return to La Jolla in the warm summer months.

4.2. Aggregation behavior

Because shovelnose guitarfish and bat rays had high and moderate levels of site fidelity to the lee of La Jolla Submarine Canyon, behavioral thermoregulation with regard to gestation may partly explain their seasonal presence. This area was hypothesized to be a gestating ground for leopard sharks due to the divergence zone created by the canyon's bathymetry, which results in calmer, warmer water (Nosal et al. 2013). Warm conditions are hypothesized to accelerate embryonic growth in pregnant ectothermic elasmobranchs (Hight & Lowe 2007, Jirik & Lowe 2012). As SST increases generally throughout the SCB due to climate change, elasmobranchs may become less concentrated at the aggregation site, at least during the peak of summer, because they will be able to access similarly warm water over a greater horizontal and vertical range (i.e. the aggregation site will be less special in terms of warm water temperature). This trend may be mitigated by other theoretical social benefits of aggregating (e.g. heightened vigilance), which have not been thoroughly examined.

Around La Jolla, both shovelnose guitarfish and bat rays were detected most often over soft substratum habitat. Given that the head of the La Jolla Submarine Canyon primarily consists of soft substratum, it is possible that the apparent selection of soft substratum could actually be the selection of the warmer water temperatures in this area. However, shovelnose guitarfish affinity to soft substratum, independent of water temperature, would be consistent with a past visual observation study done around Santa Catalina Island (Meese & Lowe 2019). Shovelnose guitarfish also had a much stronger selection for soft substrate than bat rays, which could be explained in part by coloration, with bat rays (darker) having improved crypsis over darker hard substrate such as rocky reefs and shovelnose guitarfish (lighter) having improved crypsis over soft substrate such as sand. It may also be explained by foraging differences between species. Adult shovelnose guitarfish eat sediment-burying fish (*Batrachoididae* spp.), crustaceans (*Brachyura* spp., *Stomatopoda* spp.), and squid (*Loliginidae* spp.), which may be more commonly associated with soft sediments, whereas bat rays were observed to forage for shellfish over hard substrate in La Jolla (Limbaugh 1955, Valenzuela-Quiñonez et al. 2018).

4.3. Conservation implications

This study is the first to demonstrate seasonal philopatry (≥3 consecutive summers) in shovelnose guitarfish and California bat rays to La Jolla, California, at an aggregation site that happens to be within the small no-take Matlahuayl SMR. Here, large, possibly pregnant, females were most abundant likely due to its favorable warm, calm waters (Fig. 4A). The Matlahuayl SMR, formerly the San Diego-La Jolla Ecological Reserve, is one of the oldest no-take reserves in southern California. It was established in 1971 to protect 2 vulnerable populations, the sessile green abalone and the highly residential vermillion rockfish (Parnell et al. 2005). Although it was not originally intended to enhance populations due to its small size, the Matlahuayl SMR completely encompasses the aggregation site of multiple elasmobranchs, including the leopard shark (Nosal et al. 2013), soupfin shark (Nosal et al. 2021), and, as demonstrated by this study, the shovelnose guitarfish and California bat ray. Without this MPA, all of these species would be vulnerable to group capture, which was reported historically during the elasmobranch derbies at

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known aggregation sites in Elkhorn Slough and Humboldt Bay (Herald et al. 1960, Carlisle et al. 2007). This study demonstrates the potential value of small no-take reserves as one of many management tools for the conservation of coastal elasmobranch species, specifically the shovelnose guitarfish and bat ray, should their populations decline significantly, and describes environmental preferences that could help in informing future MPA placement. Although fishing pressure on both species is not presently a serious threat in California, predictable coastal aggregations with pregnant females could be targeted in the future. Identifying and monitoring such aggregation sites is essential to maintaining and monitoring stable populations; disregarding these aggregation sites of pregnant females could lead to swift population declines, as was the case for the now Critically Endangered Brazilian guitarfish *Pseudobatos horkelii*. In this species, pregnant females were heavily targeted in the summer months by artisanal fisheries along the coast of Brazil, which led to an estimated 85% decline in abundance from 1975 to 1990 (Casselberry & Carlson 2015). Thus, it is important to identify the environmental, temporal, and spatial conditions in which bat rays and shovelnose guitarfish aggregate, especially when aggregations are composed primarily of pregnant females, to proactively monitor and understand population status.

Although landings in California of shovelnose guitarfish and California bat rays are low, both species are taken as targeted and incidental catch in the Mexican artisanal elasmobranch fishery. Shovelnose guitarfish and bat ray landings are prominently high in fisheries along the Pacific coast of Baja California (comprising 46 and 10% of all elasmobranchs caught between 2006 and 2008, respectively; Cartamil et al. 2011) and Baja California Sur (28 and 9% of all elasmobranchs caught between 2000 and 2010, respectively; Ramirez-Amaro et al. 2013). As of 2014, bat rays are classified as Least Concern by the IUCN (van Hees et al. 2015). On the other hand, shovelnose guitarfish are classified as Near Threatened by the IUCN due to their slow growth, frequent capture in artisanal gillnet fisheries, and susceptibility to bycatch by demersal shrimp trawl fisheries in Mexico (Farrugia et al. 2016). Shovelnose guitarfish fisheries remain underregulated, with a lack of species-specific management efforts in both the USA and Mexico. Although this study could not determine the extent of cross-border migration of either species due to the sparse coverage of acoustic receivers in Mexico, it is highly likely that movement was occurring between the 2 countries due to La Jolla's proximity to the border as well as a lack of a biogeographic boundary between the 2 countries. Shovelnose guitarfish movement from southern California into Mexico is supported by genetic evidence, which demonstrated a singular subpopulation from Point Conception, California, to Punta Eugenia, Baja California, excluding the Channel Islands (Meyer 2020). Only 1 tagged bat ray was detected in Mexico, but most tagged bat rays were not detected outside of La Jolla outside of the summer months. The reason for this lack of detections could be that the bat rays occupied deeper offshore habitats or migrated south towards Mexico. There is a seasonal elasmobranch fisheries closure on the Pacific coast of Baja California from 31 May to 31 July, which should provide some protection for both species during the summer months, when individuals are most likely to aggregate. However, elasmobranch bycatch is still prevalent during this closure, and there is no species-specific action, such as daily bag limits, in place for either species in Mexico (van Hees et al. 2015, Farrugia et al. 2016). Identifying and monitoring aggregation sites in Mexico for more location-based conservation will greatly benefit both species in areas with high fishing pressure.

For decades, researchers have recognized the vulnerability of bony fish aggregations to overexploitation and fishery collapse (Coleman et al. 1996, Sadovy de Mitcheson & Erisman 2012), leading to protective measures at the time and place of aggregation behavior to mitigate population declines and restore fisheries catch (Erisman et al. 2017, Waterhouse et al. 2020). Because elasmobranch fishes are widely known to aggregate for reproduction purposes such as gestation, birth, and mating (Papastamatiou & Lowe 2012) and are vulnerable to such fishing pressures, similar protective measures should be considered. The establishment of MPAs has been successful in protecting various elasmobranch species, such as the juvenile pigeye shark *Carcharhinus amboinensis* and spot-tail shark *C. sorrah* within the Great Barrier Reef Marine Park, Australia (Knip et al. 2012), and the short-tailed ray *Dasyatis brevicaudata* within the Poor Knights Islands Marine Reserve, New Zealand (Le Port et al. 2012). Understanding elasmobranch migration, philopatry, and aggregation is integral to proper elasmobranch conservation; thus, further studies should continue to identify aggregation sites and migration pathways.

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